Handling Code-Switching in Arabic-English Tasks

Benchmarking LLMs on Arabizi and Mixed Arabic-English Text

Yousef Joudeh · Seyed Almadani
Abdulaziz Mohammed · Abdallah Almekhyal
CS496 · Department of Computer Science · Kuwait University

Supervised by
Dr. Lulwah Alkulaib

Outline

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2. Datasets & Annotation
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4. Results & Gold-Label Discovery
5. Error Analysis & Discussion
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Why Code-Switching Matters

The Phenomenon

Arabic speakers on social media routinely mix:

- Arabic script + English
- Arabizi (Latin-script Arabic with numerals:
- 3 = ع, 7 = , ح)
- Other languages (e.g., French)

Example:

"ana raye7 el jam3a today"

The Gap

- LLMs are largely untested on Arabic-English code-switching
- Prior work focused on translation or encoder-based models only
- Arabizi adds complexity most benchmarks ignore
- Limited prior evaluation of recent LLMs on script-aware token-level LID

Research Questions & Hypotheses

RQ1: How accurately can LLMs perform token-level language identification on code-switched text?

H1: AR-LAT (Arabizi) tokens will have lower F1 than AR and EN tokens
H2: Few-shot prompting will yield higher macro F1 than zero-shot
H3: Models will perform differently across the two datasets

RQ2: How do different LLMs compare on Arabic-English token-level LID?

H1: Models differ significantly in token-level accuracy; macro F1 is used for ranking
H2: The best model on AR will also rank first on AR-LAT

Datasets

Statistic	Arabizi (WANLP 2022)	AR-EN CS (Kaggle 2021)
Sentences	2,574	2,507
Tokens	29,810	30,321
Avg. tokens/sent	11.6	12.1
AR tokens (%)	9.0%	63.9%
EN tokens (%)	55.6%	23.4%
AR-LAT tokens (%)	21.0%	—
Source	Twitter / Reddit	WhatsApp / FB / Twitter

Why These Two?

Complementary coverage:

- Arabizi corpus = English-dominant, Latin-script mixing
- AR-EN CS corpus = Arabic-dominant, script-based mixing

Combined:

- 5,081 sentences
- 60,131 tokens

Note: AR-EN CS originally has no AR-LAT label, script inconsistency is discussed later.

Unified 5-Label Scheme

AR	EN	AR-LAT	OTHER	OL
Arabic script	English	Arabizi	Punct/symbols	Other lang.

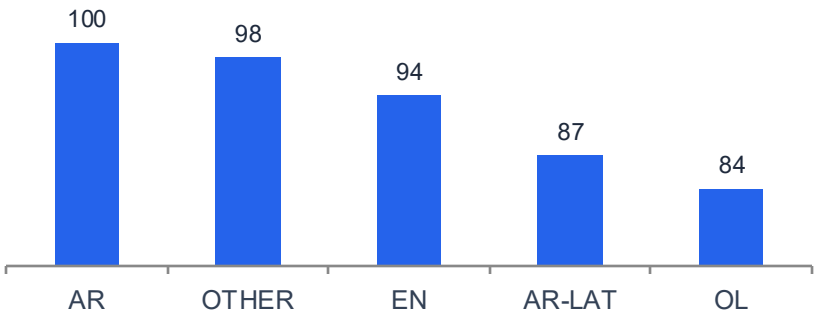
Annotation & Inter-Annotator Agreement

3 annotators independently labeled 250 code-switched sentences (2,952 tokens) from the Arabizi dataset

- Stratified to cover all label categories
- Excluded purely monolingual English sentences
- No discussion between annotators

Metric	κ	Interpretation
Ann. 1 vs 2 (Cohen)	0.9873	Almost perfect
Ann. 1 vs 3 (Cohen)	0.9823	Almost perfect
Ann. 2 vs 3 (Cohen)	0.9848	Almost perfect
Fleiss' κ (overall)	**0.9847**	**Almost perfect**

Per-Label Agreement Rates



35

of 2,952 tokens with any disagreement (1.2%)

Well above 0.80 reliability threshold

Experimental Setup

GPT o4-mini

OpenAI — o-series reasoning model

Claude Sonnet 4.6

Anthropic — First CS eval for this model

Gemini 2.5 Flash

Google — Optimized for speed

LLaMA 3.3 70B

Meta (open-source) — Via Together AI API

Fixed Parameters

Parameter	Value
Temperature	0.0
Max tokens	2,048
Sample size	500 (250/dataset)
Random seed	42
Few-shot examples	5 (all 5 labels)

Prompting Strategies

Zero-shot: Task instructions + label definitions + JSON format + compliance rules

Few-shot: Same as zero-shot + 5 worked examples covering all 5 labels (no overlap with eval set)

Evaluation Metrics

Primary Metric

Macro F1

Unweighted average F1 across all 5 labels — gives equal importance to rare labels like OL and AR-LAT, avoiding inflated scores from dominant classes.

Token-Level Accuracy

Overall % of correctly labeled tokens — simple global measure

Micro F1

F1 weighted by label frequency — global performance summary

Per-Label P / R / F1

Precision, recall, F1 for each of the 5 labels separately

Parse Failure Rate

% of sentences where model output was unparseable JSON

Interpretability Tools

5 × 5 Confusion Matrix

Reveals which labels are systematically confused (e.g., AR-LAT ↔ EN, AR → AR-LAT). Essential for identifying error patterns beyond aggregate scores.

Qualitative Error Analysis

Categorized error taxonomy with contrastive examples showing where models agree and disagree on ambiguous tokens.

Why this set? Accuracy and F1 are standard for token-level LID. Macro F1 prevents high-frequency labels (EN, OTHER) from masking poor performance on rare but important labels (AR-LAT, OL). Parse failure rate is non-standard but essential for LLM evaluation — invalid outputs can't be scored at all.

Results: Overall Performance

Model	Strategy	Accuracy	Macro F1	PF%
GPT o4-mini	Zero-shot	0.754	0.621	7.0
GPT o4-mini	Few-shot	0.734	0.649	13.0
Claude Sonnet	Zero-shot	0.737	**0.683**	9.2
Claude Sonnet	Few-shot	0.737	0.682	8.4
Gemini Flash	Zero-shot	0.760	0.680	7.6
Gemini Flash	Few-shot	**0.763**	0.661	7.6
LLaMA 70B	Zero-shot	0.742	0.618	11.4
LLaMA 70B	Few-shot	0.750	0.608	13.2

Highest Macro F1

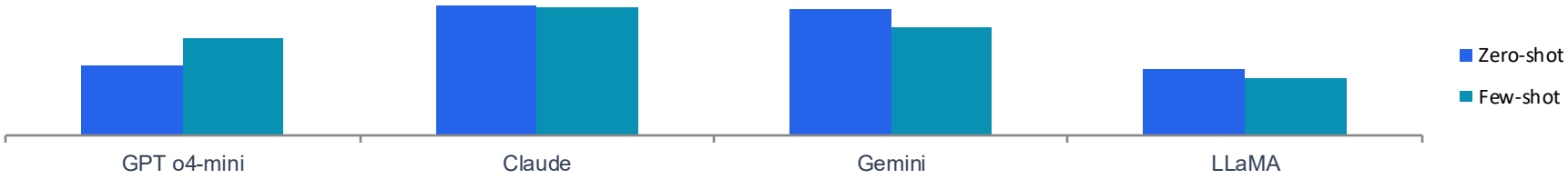
Claude Sonnet ZS: 0.683

Highest Accuracy

Gemini Flash FS: 0.763

Few-shot ≠ better

Parse failures nearly double for GPT o4-mini (7% → 13%)



Results: Per-Label F1 Breakdown

Model	Str.	AR	EN	AR-LAT	OTHER	OL
GPT o4-mini	ZS	.568	.942	.465	.908	.222
GPT o4-mini	FS	.576	.938	.467	.909	.357
Claude	ZS	.574	.934	.470	.915	.522
Claude	FS	.566	.939	.468	.914	.522
Gemini	ZS	.598	.941	.471	.912	.480
Gemini	FS	**.609**	.943	.460	.922	.370
LLaMA	ZS	.516	.935	.468	.887	.286
LLaMA	FS	.558	.938	.485	.884	.174

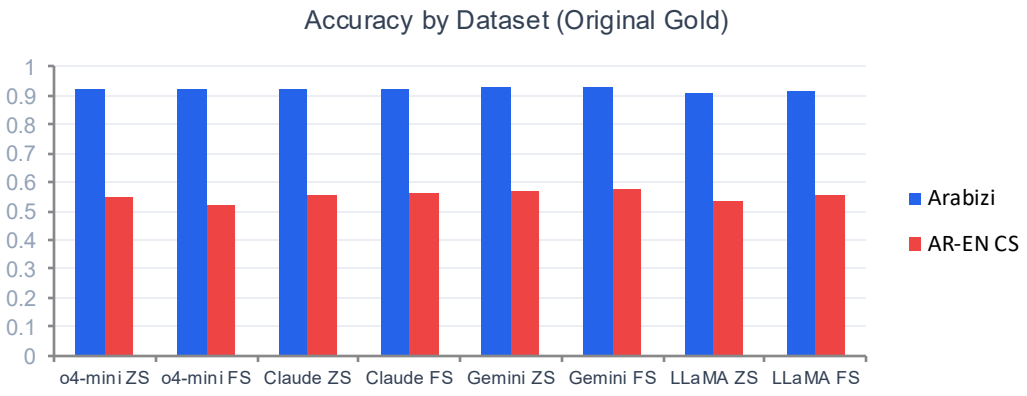
H1 & H3 Supported

AR-LAT F1 (0.46–0.49) far below EN (0.93–0.94). Arabizi is the hardest label for every model.

H2 (RQ2) Not Supported

Gemini ranks 1st on AR F1, but LLaMA FS is highest on AR-LAT. Script recognition ≠ Arabizi recognition.

The 35-Point Gap & Gold-Label Discovery



Key Discovery

55.1% of AR-labeled tokens in the AR-EN CS corpus are actually written in Latin script

Original corpus labeled by language identity, not orthographic script

When models labeled these as AR-LAT, they were following instructions correctly!

Corrected Evaluation (re-scored, no new API calls)

		Original		Corrected	
Model	**Str.**	**Acc.**	**MaF1**	**Acc.**	**MaF1**
GPT o4-mini	FS	.734	.649	**.940**	**.824**
Claude Sonnet	FS	.737	.682	**.944**	**.861**
Gemini Flash	ZS	.760	.680	**.942**	**.850**
LLaMA 70B	ZS	.742	.618	**.923**	**.795**

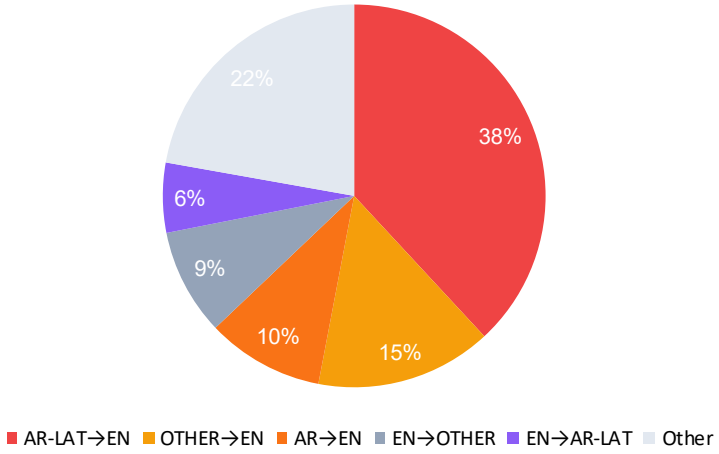
+19pp

avg. accuracy gain
after gold-label correction

Error Analysis

After Gold-Label Correction (2,320 errors remain)

Error Category	Count	%	Example
AR-LAT → EN	885	**38.1%**	La, ana
OTHER → EN	345	14.9%	AHAHHA
AR → EN	229	9.9%	motivatonال
EN → OTHER	208	9.0%	24hrs
EN → AR-LAT	137	5.9%	had
Other errors	516	22.2%	Mixed



Contrastive Examples (116 tokens with model disagreement)

Token	Gold	GPT o4	Claude	Gemini	LLaMA
"El"	AR-LAT	OL ✗	AR-LAT ✓	AR-LAT ✓	EN ✗
"Allah"	AR-LAT	AR-LAT ✓	EN ✗	EN ✗	EN ✗

Statistical Significance & Ablation

McNemar's Test: Zero-Shot vs. Few-Shot

Model	p-value	Sig.
GPT o4-mini	**0.004**	**
Claude	1.000	ns
Gemini	0.136	ns
LLaMA	0.076	ns

Pairwise Model Comparison

Comparison	p-value	Sig.
o4-mini vs Claude	0.137	ns
o4-mini vs Gemini	**0.013**	*
o4-mini vs LLaMA	**<0.001**	***
Claude vs Gemini	0.658	ns
Claude vs LLaMA	**<0.001**	***

Three Tiers Emerge:

- Tier 1: Claude ≈ GPT o4-mini
- Tier 2: Gemini
- Tier 3: LLaMA

Ablation: Targeted 10-Shot (Claude Sonnet)

Condition	Acc.	Macro F1	Parse Failures
Zero-shot	0.941	0.859	46
Few-shot (5 ex.)	0.944	0.861	42
Ablation (10 ex.)	0.946	0.858	**21**

Parse failures halved (46 → 21)

But macro F1 unchanged — short-token ambiguity (la, ana, en) is intrinsic, not a prompting problem

Hypothesis Testing Summary

RQ1-H1 AR-LAT F1 < AR/EN F1	**Supported**
RQ1-H2 Few-shot > zero-shot	**Not Supported**
RQ1-H3 Dataset-level performance differs	**Not Supported (After Correction)**
RQ2-H1 Meaningful cross-model differences	**Partially Supported**
RQ2-H2 Best AR model = best AR-LAT model	**Supported**

Limitations & Future Work

Limitations

Sample size: 500 sentences ≈ 10% of combined dataset; may not capture full CS distribution

Parse failures: 7–13% across runs, not missing at random — complex CS sentences more likely to fail

Social media only: Both corpora are informal registers; no formal or spoken code-switching

Single-model ablation: Targeted 10-shot tested only on Claude Sonnet; may not generalize

OL label sparsity: Only 6 gold OL tokens in eval set — F1 estimates unreliable

No fine-tuned baselines: Only zero/few-shot LLMs tested; mBERT/XLM-R may differ

Future Directions

1. **Audit other CS benchmarks**
   Language-vs-script labeling may affect other corpora combining different annotation conventions

2. **Chain-of-thought prompting**
   Require models to reason about surrounding tokens before labeling — may resolve short-token ambiguity

3. **Fine-tuned sequence labelers**
   mBERT / XLM-RoBERTa with contextual objectives for comparison with LLM prompting

4. **Optimize few-shot composition**
   Systematically vary number and type of examples to find optimal format compliance

Key Takeaways

1. **Annotation Conventions Matter**

   Gold-label inconsistency inflated error rates by 19pp. Language-vs-script labeling is a systemic risk in code-switching evaluation.

2. **LLMs Are Better Than They Look**

   After correction, all models achieve 92–95% accuracy. Claude Sonnet ranks first (macro F1 = 0.861).

3. **Few-Shot Isn't Always Better**

   McNemar's test: significant improvement for only 1 of 4 models. Token-level LID is sufficiently defined by label definitions alone.

4. **Data > Models**

   The 35-point dataset gap dwarfs the 7.5-point model spread. Dataset composition is a stronger predictor of difficulty than model choice.

5. **Short-Token Ambiguity Is Intrinsic**

   Tokens like la, ana, en remain misclassified even with targeted prompting. Needs contextual sequence labeling.

Questions & Discussion

Thank you for your attention

github.com/abojoudah/cs496-codeswitching

Yousef Joudeh · Seyed Almadani · Abdulaziz Mohammed · Abdallah Almekhyal
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